When Search Is Not Enough: Diagnosing Failure Before Constructing Verified Reasoning Capabilities

Chimdi Walter Ndubuisi University of Missouri

CORA on ARC-AGI-2 · development results · ARC Prize Research Summit, 23 October 2026

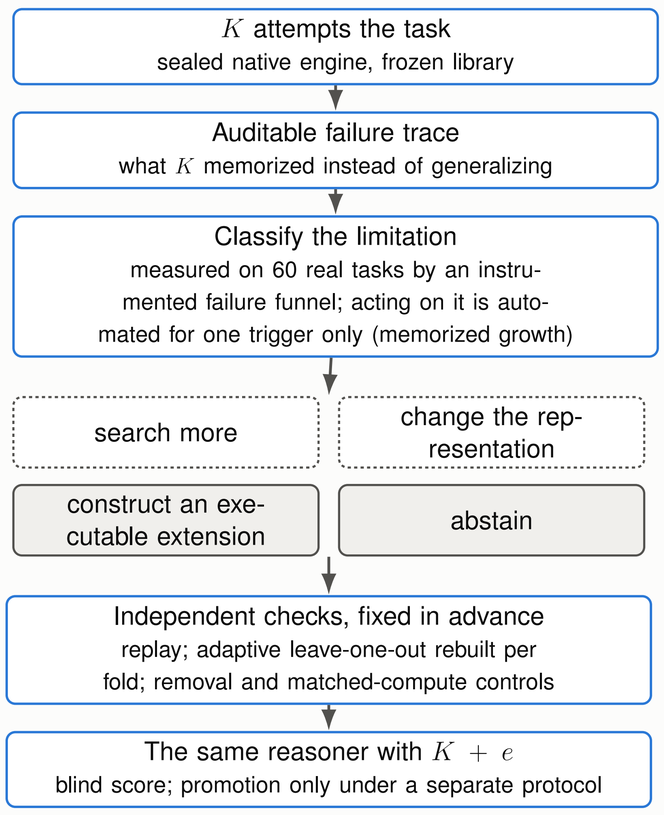
The question

How can a reasoning system know when its existing way of solving problems is inadequate, and acquire a better one without compromising the evidence used to judge it?

A failed solve can need different responses:

- **search correction**: the answer is expressible, search missed it
- **representation correction**: the needed view was screened out
- **executable capability construction**: the language lacks it
- **evidence insufficiency**: the examples underdetermine the rule
- **certification failure**: a valid candidate fails the checks

The CORA loop

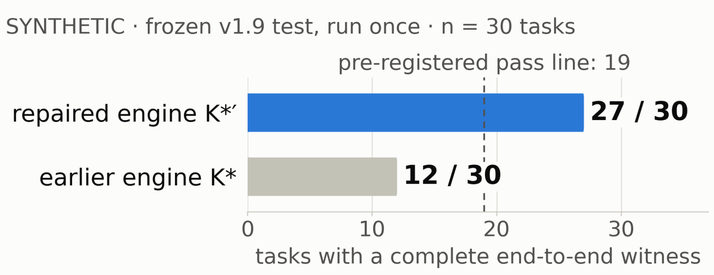


Solid: implemented and measured. Dashed: proposed, not automated. A general failure-type controller is **proposed, not implemented**.

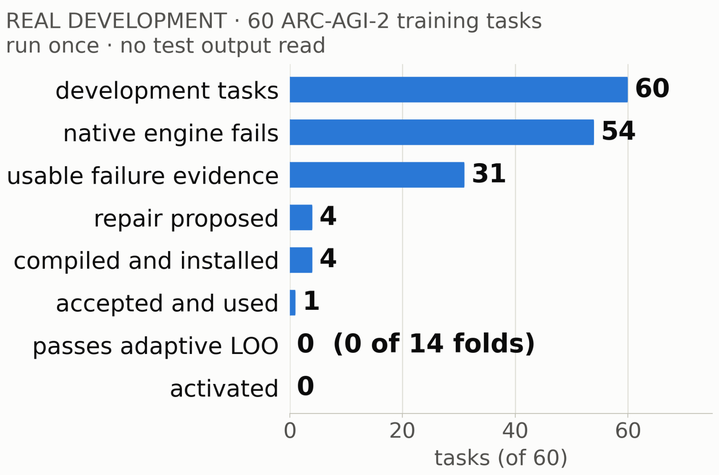
Four claims, four kinds of evidence

(a) the extension causes the solve	development only: nothing accepted without it, or with matched compute; not certified
(b) the real failure trace matters	demonstrations-only and shuffled failures never gave an accepted, used extension (4 tasks)
(c) a bounded semantic extension	**not established**
(d) retention and later reuse	**untested**

What has been measured

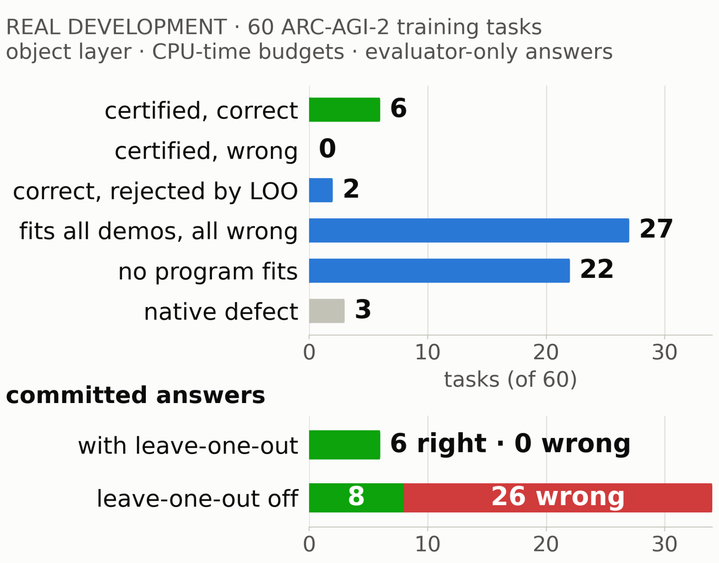


A synthetic claim only. The repaired engine also accepted every deliberately wrong extension it was given (5 of 5, 28 of 28).



One constructed extension was accepted and used by the unchanged engine; none survived adaptive leave-one-out. **No new ARC solve**: the certified training score stays 185/1,000.

Where CORA fails on real tasks



The checks are not the problem: nothing wrong was certified, and switching leave-one-out off would add 2 right answers and 26 wrong ones. **The rules are**: in the 27 "fits all demos" tasks the programs memorize. 21 of 25 use constant parameters, mostly a grown cell pattern keyed to a colour or an exact shape, and each leave-one-out fold rebuilds a different one (59 of 74 failed folds).

- Native's own relational re-search finds no program (0 of 11).
- Three times the search budget fixes none (0 of 14).
- Given the true answer as a demonstration (diagnostic only), it fits 26 of 27, in the same view in 24, by memorizing one more case.
- The view-admission failure we first traced explains only 7 of 76 failed folds.

Frozen neural guides did not pick views better than the fixed order (Qwen2.5-1.5B 19 of 35, I-JEPA 3 of 35; pre-registered, synthetic).

Repairs tested before use

- Forcing a view to be searched first: no task gained, 5 of 24 synthetic tasks harmed (4 correct answers lost, 1 false acceptance). **Not run.**
- Fold-local removal of duplicate views: gained 1, harmed 4 of 41. **Rejected.**
- Every fold chooses its representation from **its own demonstrations only**; anchoring a fold to the parent's view is forensic, never a certificate.

Competition status, 14 October

OFFICIAL KAGGLE · native CORA v23: **0.00**.

NEURAL_TTT_BASELINE · a reproduction of the public NVARC Qwen3-4B test-time-training pipeline, **not CORA**: submitted 14 October, **score pending**. On the 240-task stand-in it decoded every task (259 of 259 genuine predictions, 0 fallbacks) in 4.3 h.

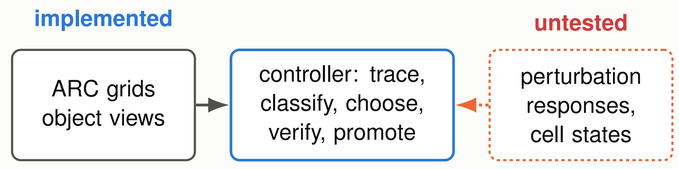
DEVELOPMENT · whole native harness: 12 of 60; 18 of its 29 claimed answers were wrong, because the pipeline layer answers first. Reordering the answers it already produces (shadow study): 14 of 60, and 36 to 39 of 60 on synthetic C2-dev, none lost. Engineering only; not submitted.

Next: construct the missing operation PROPOSED

- **Trigger**, from demonstrations only: a memorized constant growth pattern that leave-one-out cannot rebuild.
- **Construct** a derived pattern operator from a declared, bounded grammar. CORA's 9 growth families cover only 5 of the 27, so the grammar must grow, and be frozen first.
- **Verify**: native re-induction in every fold, full adaptive leave-one-out, blind scoring.
- **Controls**: matched extra search, failure-blind and shuffled-failure construction.

Not yet run. A gain would be compositional growth, not a new primitive.

A portable controller? BIOLOGICAL APPLICATION: UNTESTED



The hypothesis: the controller and its evidence and certificate interfaces are portable; ARC operators are not. A test needs typed adapters, training-only diagnosis, held-out perturbations and cell states, and confounder controls. No biological data or result exists here.

Not claimed · evidence

No CORA solve or score gain on ARC-AGI-2; no semantic invention; no automatic failure-type controller; the neural score, when it arrives, is not a CORA result; no domain generality; no results from the separate VDCG project. The 120-task pilot is frozen and **has not been run**.

Every number, with commit and hash: `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md` and `records/CORA_FAILURE_FUNNEL_20261014.md`; figure hashes: `figures/PROVENANCE.json`.